

Abstract

We report the first completed exercise of a re-formalization program: a pat framework observation (“composite numbers are multiple projection superpositions of primes”) is re-stated in pure known mathematics and formally proved in Lean 4 / mathlib with *no* pat framework concepts. The re-statement is the fundamental theorem of arithmetic; the proof is four theorems: existence of the prime factorization (`prod_factorization_pow_eq_self`), primality of the factorization support, uniqueness of the factorization (via mathlib’s `factorizationEquiv`), and the existence of a nontrivial prime factor for every composite number (`exists_prime_and_dvd`). All four compile with zero errors against mathlib v4.32.2. The exercise demonstrates the program’s method: walk the genealogy map from the pat concept to its standard-mathematics coordinate, formalize the equivalence on the way, and keep the pat vocabulary out of the proof.

Lean, formalization, arithmetic, fundamental theorem of arithmetic, Opat, reproducibility

Composite Numbers as Prime Factorization: A Pure-Known-Mathematics Re-formalization of a Pat Observation

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1 Introduction

Program. This paper is the first completed exercise of a re-formalization program (“0pat”): take the exercises of the pat framework — a body of Lean-formalized observations about basepoint- relative operator semantics — and re-prove their mathematical content using *pure known mathematics*: only mathlib’s existing definitions and theorems, no pat framework concepts (basepoints, phase locking, interlock pairs, projection superpositions).

Method. Each exercise is located on a genealogy map (rows: pat concepts R1–R25; columns: mathematical genealogies — geometry, algebra, analytic number theory, topology, category theory, representation, physics, construction). The map’s entries are the paths: a pat concept at coordinate (R_i, C_j) has a standard-mathematics correspondent at that coordinate. The re-formalization walks the path: re-state the pat claim at its coordinate in standard mathematics, formalize the equivalence, and prove. Since the mathematical claim is known (the “answer” exists in mathlib), the work is path-finding, not theorem-finding.

This exercise. The pat observation “composite numbers are multiple projection superpositions of primes” sits at $(R23, C3)$ on the genealogy map (prime circle $\times$ analytic number theory). Its standard- mathematics correspondent is the fundamental theorem of arithmetic. We re-state and prove it.

2 Re-statement (0pat)

Pat claim. A composite number is a superposition of prime projections: its prime structure is fully determined by the primes that divide it and their multiplicities.

0pat re-statement (fundamental theorem of arithmetic).

1. *Existence:* every nonzero natural number equals the product of its prime powers: $\prod_p p^{e_p} = n$ where $e_p = \text{factorization}(n)(p)$.

2. *Primality of support*: $e_p > 0$ only for prime p .
3. *Uniqueness*: any prime-supported factorization whose product equals n is exactly n 's factorization.
4. *Composite structure*: every composite $n > 1$ has a prime factor $p < n$.

3 Formalization

The proof is a single Lean file (`proof.lean`, 90 lines), importing only `Mathlib.Data.Nat.Factorization.Basic` and `Mathlib.Data.Nat.Basic`. Four theorems:

Theorem	Statement	mathlib anchor
<code>factorization_represents</code>	$n \neq 0 \Rightarrow \prod_p p^{e_p} = n$	<code>Nat.prod_factorization</code>
<code>factorization_support_prime</code>	$p \in \text{supp } e \Rightarrow \text{prime } p$	<code>Nat.factorization_support_prime</code>
<code>factorization_unique</code>	$\text{prime support} + \text{product} = n \Rightarrow f = n.\text{factorization}$	<code>Nat.factorization_unique</code>
<code>composite_has_prime_factor</code>	$1 < n \wedge \neg \text{prime } n \Rightarrow \exists p, \text{prime } p, p \mid n, p < n$	<code>Nat.exists_prime_factor</code>

Uniqueness via the factorization equivalence. The cleanest route to uniqueness is `mathlib`'s `factorizationEquiv` : $\mathbb{N}^+ \simeq \{f : \mathbb{N} \rightarrow_0 \mathbb{N} \mid \forall p \in \text{supp } f, \text{prime } p\}$: the inverse maps f to $f.\text{prod}(\cdot)$ (a `mathlib` theorem), so a prime-supported f with product n is the preimage of n under the inverse — hence equals `factorizationEquiv(n)`, which is definitionally $\langle n.\text{factorization}, \cdot \rangle$. The whole proof is: $f = \text{factorizationEquiv}(n) = n.\text{factorization}$.

Verification. Compiled with Lean 4.32.2 (`mathlib` v4.32.2), zero errors, zero `sorry`, zero axioms beyond `mathlib`'s. Build command in the artifact.

4 Relation to the pat exercise

The original exercise (`pat_pnp` exercise XXIII, claim R234) formalized the same observation inside the `pat` framework, using `pat` vocabulary. This re-formalization shows that the mathematical content survives the removal of the framework: the observation is the fundamental theorem of arithmetic, and its proof is entirely within known mathematics. The `pat` vocabulary is preserved in the exercise record (README) as provenance, not in the proof.

5 Conclusion

The first `0pat` exercise is complete: composite numbers as prime factorization, proved with pure known mathematics in four theorems, zero errors. The method (genealogy-map path-walking, formalize the equivalence on the way) is validated on this exercise and is the template for the remaining ones.